LEVEL THE CURVE

ENGINEERING INTERNSHIP PILOT PROGRAM

Assistive Device Design & Adaptive Technology

Access Independence Dignity Designing for All.

Organization:	Level The Curve Inc. **All dates/sections subject to change**
Duration:	10 Weeks (June 2 – August 8)
Schedule:	Monday – Friday, 9:00 AM – 5:00 PM
Demo Day:	Friday, August 8 Capstone Showcase & Product Pitch
Location:	Level The Curve Design Lab, New York, NY
Mentors:	LTC Co-Founders (Stefan, Eli, Khan) + Licensed P.E. Advisors
Cohort Size:	6–10 Interns (Teams of 2–3)
Affiliates:	NYU Ability Project | El Barrio | Mount Sinai Hospital

About Level The Curve

Level The Curve was founded by Stefan (CEO), Eli (COO), and Khan (Chief Engineer) three individuals united by personal experience with spinal cord injury and a shared engineering vision. The company designs **affordable, sleek, consumer oriented assistive devices** that empower people with disabilities to perform Activities of Daily Living (ADLs) with independence and dignity. LTC's flagship ***EZ Adapt*** product line, including the Vesta eating utensil holder, demonstrates the company's commitment to products that are functional, discreet, and accessible at a low price point.

This internship program extends that mission by bringing in the next generation of engineers to design, prototype, and validate new assistive devices guided by the same lived experience philosophy that drives everything LTC builds.

Program Objectives

Upon completion of this pilot program, each intern will have demonstrated competency in the following:

1. Identify a real world accessibility challenge faced by individuals with disabilities in one of five ADL categories

2. Design an adaptive device solution validated through engineering analysis (FEA, ergonomic modeling, material selection)
3. Collect direct feedback from end users with disabilities, caregivers, and occupational therapists through structured user testing
4. Build a functional, affordable prototype using 3D printing, CAD/CAM, and rapid fabrication methods
5. Pitch the final product with a viable path to manufacturing, pricing at a consumer friendly level consistent with LTC's affordability mission

The LTC Engineering Design Process

This program is structured around LTC's core philosophy: ***design with the user, not just for the user***. Intern teams will move from problem discovery through analysis, prototyping, and validation working directly alongside individuals with disabilities at every stage. Through *customer discovery*, *agile development*, and *iterative design*, interns will ensure their solutions meet the three pillars of every LTC product: **Affordability, Equity, and Empowerment**. All engineering work will be reviewed by Khan (Chief Engineer) and a supervising P.E. advisor.

Available ADL Focus Categories

Each intern team will select one of the following categories, aligned with Activities of Daily Living where assistive technology has the greatest impact. Projects will be scoped to address a documented, real world need within the chosen domain:

Eating & Feeding	Dressing & Grooming	Mobility & Transfer	Home & Environment	Communication & Recreation
Utensil grips, adaptive cups, plate guards, one-handed food prep tools	Button hooks, zipper pulls, sock aids, adapted personal care tools	Transfer boards, wheelchair accessories, grab bar systems, reach extenders	Adapted switches, door/cabinet handles, key turners, environmental controls	Pen/stylus grips, phone holders, adaptive gaming, page turners

Program Phases (10-Week Timeline)(Subject to change)

Phase 1: Research & Problem Discovery (Weeks 1–2)

Interns immerse themselves in the disability and assistive technology landscape. Teams research their chosen ADL category, review existing products on the market, identify gaps in affordability or functionality, and most critically conduct discovery interviews with end users. LTC's affiliate network (NYU Ability Project, Mount Sinai, El Barrio) provides direct access to communities and clinicians.

- Conduct user interviews with individuals with disabilities and caregivers
- Shadow occupational therapists to observe ADL challenges firsthand
- Perform competitive analysis of existing assistive devices (cost, function, aesthetics)
- Review relevant standards: ADA guidelines, FDA Class I device requirements, ASTM F3131

Deliverable: Problem Discovery Report with user personas, needs assessment, and market gap analysis

Phase 2: Conceptual Design & Engineering Analysis (Weeks 3–4)

Teams develop 2–3 conceptual device solutions grounded in engineering fundamentals. Each concept is evaluated for structural integrity (FEA), ergonomic fit (anthropometric data), material biocompatibility, and critical cost to manufacture at LTC's price targets. A decision matrix weighted toward affordability, safety, and user dignity guides concept selection.

- Create parametric CAD models (SolidWorks, Fusion 360) with design for manufacturing intent
- Perform FEA stress analysis for grip forces, drop scenarios, and fatigue loading
- Select materials balancing biocompatibility, durability, and cost (ABS, PETG, TPU, silicone)
- Ergonomic analysis using anthropometric data for limited dexterity users

Deliverable: Preliminary Design Package with CAD models, analysis results, BOM, and cost estimate

Phase 3: User Feedback & Design Review (Weeks 5–6)

Following LTC's founding principle, lived experience drives design interns present preliminary concepts directly to end users, caregivers, occupational therapists, and the LTC co-founders. Structured usability testing with foam/3D printed mockups captures feedback on grip comfort, discretion, ease of use, and aesthetic appeal. This phase mirrors LTC's own customer development process.

- Usability testing sessions with end users through LTC affiliate network
- Formal design review with Khan (Chief Engineer) and P.E. advisor panel
- Feedback from occupational therapists on clinical viability
- Design Failure Mode & Effects Analysis (DFMEA) focused on user safety

Deliverable: Revised Design Specification incorporating all user and clinical feedback

Phase 4: Prototype Build & Iterative Testing (Weeks 7–8)

Teams fabricate functional prototypes using LTC's rapid prototyping resources. Devices are tested against real world use scenarios: grip strength ranges, repeated wash cycles, drop testing, and extended daily use. Interns iterate designs based on test data, practicing the agile development cycle that defines LTC's product workflow. Cost per unit is tracked against LTC's affordability benchmarks.

- Fabricate prototypes using FDM/SLA 3D printing, silicone molding, or manual fabrication
- Execute test plans: grip force testing, cyclic loading, drop tests, cleaning durability
- Conduct second round user testing with functional prototypes
- Track unit cost and validate against LTC price target (≤$30 retail)

Deliverable: Working Prototype with Test Report and cost per unit analysis

Phase 5: Technical Documentation & Capstone Pitch (Weeks 9–10)

Interns prepare a comprehensive technical and business package, then pitch their product to a panel including the LTC co-founders, P.E. advisors, affiliate partners, and potential end users. The pitch simulates real product launch decisions and best design practices may be considered for addition to LTC's EZ Adapt product line.

- Complete drawing package with GD&T per ASME Y14.5
- Bill of materials, manufacturing plan, and retail pricing strategy

- User manual and product safety documentation
- 15-minute capstone pitch + live product demonstration to evaluation panel

Deliverable: Final Technical Package, Business Case, and Capstone Presentation

Typical Weekly Structure

Day	Activities
Monday	Team standup, weekly sprint planning, mentor check in with LTC co-founders
Tuesday	CAD/FEA design sessions, material testing, engineering analysis
Wednesday	Workshop rotation: assistive tech deep dive, 3D printing lab, or professional development
Thursday	Prototype fabrication, user testing sessions (via affiliate network), iteration
Friday	Design review with Khan & P.E. advisor, documentation, team retrospective

Intern Deliverables Summary

Week	Deliverable	Reviewed By
2	Problem Discovery Report (user personas, needs, market gaps)	LTC Co-Founders + P.E. Advisor
4	Preliminary Design Package (CAD, FEA, BOM, cost estimate)	Khan (Chief Engineer) + Peers
6	Revised Design Spec (post user testing & clinical feedback)	LTC Team + OT Clinicians
8	Working Prototype + Test Report + Cost Analysis	Khan + P.E. Advisor + End Users
10	Final Technical Package + Business Case + Capstone Pitch	Full Evaluation Panel

LTC Mentorship & Professional Development

What sets this program apart is mentorship from founders who ***live the experience every day***. Stefan and Eli bring the invaluable perspective of end users with spinal cord injuries, while Khan provides on hand engineering leadership. Together with licensed P.E. advisors, they ensure every design meets both technical rigor and human centered design standards.

- Weekly design critiques with LTC co-founders (user perspective + engineering review)
- Guest lectures from occupational therapists, rehabilitation engineers, and disability advocates
- 3D printing and rapid prototyping certification (FDM, SLA, silicone molding)
- Professional development: engineering ethics, PE licensure path, inclusive design principles
- Site visits to affiliate partners (NYU Ability Project, Mount Sinai rehabilitation facilities)

Evaluation Criteria

Final projects will be evaluated by a panel of LTC co-founders, P.E. advisors, affiliate clinicians, and end users on the following weighted criteria:

Criterion	Weight	Assessed By
User-Centered Design (empathy, accessibility, dignity)	**25%**	LTC Co-Founders + End Users
Technical Rigor (analysis, safety, code compliance)	**25%**	Khan + P.E. Advisor
Affordability & Manufacturability (≤$30 target)	**20%**	LTC Team
Prototype Quality & Test Results	**15%**	Khan + P.E. Advisor
Presentation & Documentation Quality	**15%**	Full Panel

*Top scoring projects may be considered for inclusion in LTC's **EZ Adapt** product line, with the intern team credited as co-designers.*

Eligibility Requirements

- Currently enrolled in an ABET accredited Mechanical Engineering, Biomedical Engineering, or Industrial Design program (rising junior or senior preferred)
- Completed coursework in Mechanics of Materials, Machine Design, or Human Factors/Ergonomics
- Proficiency in at least one CAD package (SolidWorks, Fusion 360, or equivalent)
- Demonstrated interest in assistive technology, inclusive design, or disability advocacy (projects, volunteering, or coursework)
- FE (Fundamentals of Engineering) exam passed or planned is a plus but not required

Core Values Alignment

Every project in this program must embody LTC's three core values:

AFFORDABILITY	Devices must target a retail price of $30 or less. Design for low cost manufacturing methods (FDM 3D printing, injection molding, silicone casting).
EQUITY	Solutions must be designed to serve users across a range of disabilities, income levels, and living situations. No user should be priced out of independence.
EMPOWERMENT	Products must be aesthetically pleasing, discreet, and dignified. Users should feel pride, not stigma, when using the device.

This pilot program is developed in partnership with Level The Curve Inc.
All engineering deliverables are subject to review by LTC's Chief Engineer and P.E. advisory panel.

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